



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
11-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

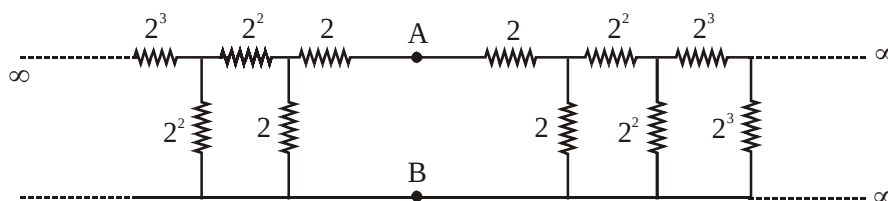
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

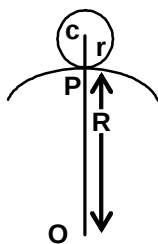
- Four charges of $6\mu\text{C}$, $2\mu\text{C}$, $-12\mu\text{C}$ and $4\mu\text{C}$ are placed at the circumference of a circle. The circle is in x-y plane and its centre at its origin. Locus of the points where electric potential is zero, is
 (A) $x = y, z = 0$ (B) $x = 0 = z$ (C) $x = 0 = y$ (D) $x = z, y = z$
- In the figure shown, the numbers indicate the values of resistance in ohm. The equivalent resistance of the combination across AB is :



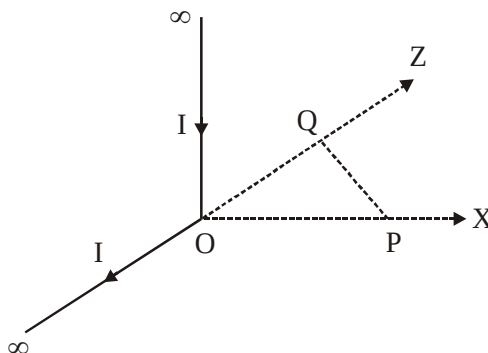
- (A) $\left(\frac{3 + \sqrt{17}}{4}\right)$ (B) $3 + \sqrt{17}$ (C) $\frac{3 + \sqrt{17}}{2}$ (D) $2(3 + \sqrt{17})$

Space for Rough Work

3. A sphere of radius r is undergoing pure rolling with constant angular velocity on a fixed curved surface of radius R with O as centre as shown. The ratio of the angle through which line PC rotates to the angle through which line OC rotates is :



- (A) $\frac{R+r}{r}$ (B) $\frac{R-r}{r}$ (C) $\frac{R+r}{R}$ (D) $\frac{R-r}{R}$
4. Figure shows combination of two semi infinite long wires kept on Z and Y-axis. It carries constant current I as shown in figure. Calculate $\int_Q^P \vec{B} \cdot d\vec{\ell}$ on line QP . Coordinate of point P and Q are $(a, 0, 0)$ and $(0, 0, a)$ respectively.



- (A) $\mu_0 I$ (B) $\frac{\mu_0 I}{2}$ (C) $\frac{\mu_0 I}{4}$ (D) $\frac{\mu_0 I}{8}$

Space for Rough Work

5. A standing wave is produced in a vapour of an element of atomic mass 254 (g mol^{-1}) at 400 K. Nodes are found to be 6.77 cm apart when frequency of source was 1000 Hz. The vapour used for the experiment is ($R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}$)
- (A) monatomic (B) diatomic (C) polyatomic (D) cannot be determined
6. The energy that should be added to an electron, to reduce its de-Broglie wavelength from $2 \times 10^{-9} \text{ m}$ to $0.5 \times 10^{-9} \text{ m}$ will be closest to :
- (A) 1.1 MeV (B) 40.56 MeV (C) 0.56 KeV (D) 5.6 eV

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
7. In a series L – C – R circuit, the voltage across resistance, capacitance and inductance are 30V each at frequency $f = f_0$.
- (A) If the inductor is short-circuited, the voltage across the capacitor will be $30\sqrt{2}$ V.
- (B) If the capacitor is short-circuited, the voltage drop across the inductor will be $\frac{30}{\sqrt{2}}$ V.
- (C) If the frequency is changed to $2f_0$, the ratio of reactance of the inductor to that of the capacitor is 4 : 1.
- (D) If the frequency is changed to $2f_0$, the ratio of the reactance of the inductor to that of the capacitor is 1 : 4.

Space for Rough Work

8. A charged particle of charge $+q$ and mass m is projected along x -axis with a speed u and enters a magnetic field at point O .

The magnetic field is confined to a region

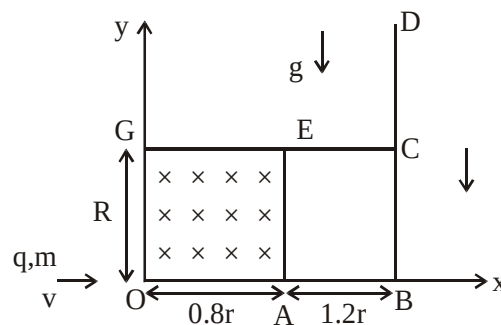
$$0 \leq x < 0.8r, 0 \leq y \leq R$$

The gravitational field is confined to a region

$$0 \leq x \leq (2r), R \leq y \leq \infty$$

The electric field is confined to a region

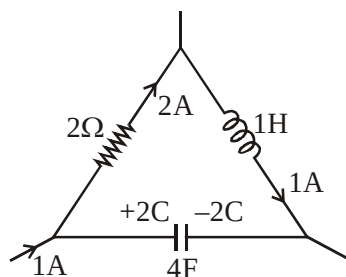
$$(2r) \leq x \leq \infty, 0 \leq y \leq \infty, R = \frac{2mu}{Bq} = 2r$$



- (A) The charged particle will not enter into gravitational field
 (B) The charged particle will not enter into electric field
 (C) The charged particle will enter into both gravitational field and electric field one after the other
 (D) The charged particle will pass through the point C with same speed u

Space for Rough Work

9. A part of a large circuit is shown. The values of resistance, inductance and capacitance are indicated here. Which of the statement(s) are correct?



- (A) The charge on the capacitor is increasing with time
 (B) The energy of the inductor is increasing with time
 (C) The energy of inductor is decreasing with time
 (D) The energy of capacitor is decreasing with time
10. A potentiometer has a cell of emf 10 V and internal resistance 2Ω . It is connected to a resistance box in which we can have any integral resistance from 1Ω to 20Ω . The resistance of the potentiometer wire is 10Ω and its length is 4.000 m. The least count of the scale is 1 mm. We wish to measure the emf of a secondary cell whose emf is 7 V.
- (A) The reading will be most accurate when we set the resistance box at 2Ω .
 (B) The reading will be most accurate when we set the resistance box at 3Ω .
 (C) If the resistance box is set at 1Ω and we balance the secondary cell on the potentiometer wire, the null point will be at 3.640 m.
 (D) If the resistance box is set at 8Ω and we balance the secondary cell on the potentiometer wire, the null point will be at 2.000 m.

Space for Rough Work

11. A bowl of water is placed on one of the plates of a beam scale, which is balanced. One of our fingers is immersed into the water, such that the bowl is not touched. Will the scale remain balanced or not, if (i) not a drop of water flows out of the bowl; (ii) the bowl was full to the brim, and the water flows off the plate of the scale?
- (A) In case (i) the scale remains balanced.
(B) In case (i) the bowl of water weighs more.
(C) In case (ii) the bowl of water weighs less.
(D) In case (ii) the bowl of water weighs the same.
12. A small ball of radius 1 cm is suspended in vacuum. It has a heater inside it which heats it at a rate of 72 W. Take Stefan's constant to be $18/\pi \times 10^{-8} \text{ W/m}^2\text{K}^4$. We have two possible paints with which we can paint the ball. One of them has an emissivity of 0.64 and another paint is black.
- (A) If the ball is painted with black paint, its surface temperature would be 1000 K.
(B) If the ball is painted with paint of emissivity 0.64, its surface temperature would be 800 K approx.
(C) If the ball is half painted with one and other half is painted with other paint, the black part will be hotter.
(D) If the ball is painted with black paint, the peak emissive power is corresponding to wavelength of 2880 nm.

Space for Rough Work

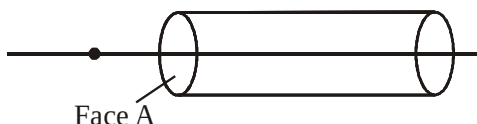
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33 , -0.30 , 30.27, -127.30 , if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25 , 5.2 then fill the bubbles as follows.

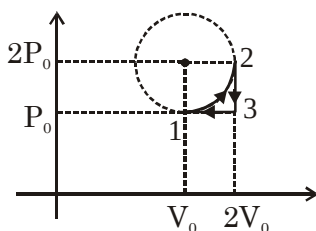
+					-				
●	●	●	●	●	●	●	●	●	
①	①	①	①	①	①	①	①	①	
②	②	②	②	②	②	②	②	②	
③	③	③	③	③	③	③	③	③	
④	④	④	④	④	④	④	④	④	
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A point source of light lies along axis, outside a long glass cylinder as shown. The refractive index of glass is $\mu = 1.5$. The distance of the source from the face A is R , where R is the radius of the cylinder. If the power of the source is $50\sqrt{2}$ W, what is the power (in W) of the beam coming from the other flat face? Neglect all partial reflections.

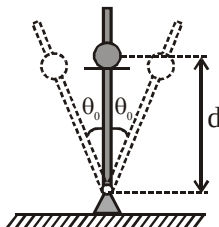


2. One mole of ideal monoatomic gas is taken along a cyclic process as shown in the figure. Process $1 \rightarrow 2$ shown is $1/4^{\text{th}}$ part of a circle as shown by dotted lines, process $2 \rightarrow 3$ is isochoric while $3 \rightarrow 1$ is isobaric. Calculate efficiency in percentage of the above cycle.



Space for Rough Work

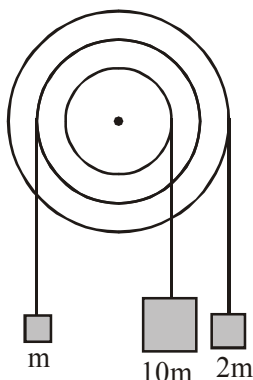
3. A bumper is mounted to a vertical rod, hinged at its bottom end, at a distance of $d = 10$ cm from the hinge. There is a small bead at rest on the bumper. The rod is started to vibrate and executes SHM about its original position with a small angular amplitude of $\theta_0 = 1^\circ$, as shown in the figure. What should the minimum frequency (in Hz) of the vibration be in order that the bead flies off? (Friction is negligible.) Take $g = 10 \text{ m/s}^2$ and $\pi^2 = 10$



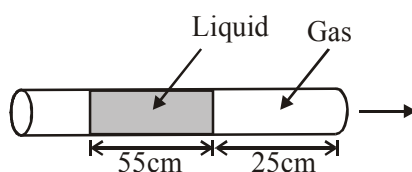
4. The radioactive carbon-14 isotope is constantly produced in the atmosphere by the cosmic rays. Nonetheless, the amount of ^{14}C on the Earth is constant, because at a height of 5-9 km, as a result of the reaction $n + ^{14}\text{N} \rightarrow ^{14}\text{C} + p$, an average of 18000; ^{14}C isotopes are produced in each second on each square metre – projected to the surface of the Earth. The half-life of the isotope is 6930 years. Estimate how many tons of ^{14}C are present on the Earth. ($R_{\text{earth}} = 6400 \text{ km}$)

Space for Rough Work

5. Three weightless pulleys with radii R , $2R$ and $3R$ are concentrically fastened together to make a single triple pulley system and mounted on an axis on which the triple pulley can rotate without friction (figure). The threads are light and they do not slip on the pulley. We suspended masses $10m$, m and $2m$ as shown. What is the acceleration (in m/s^2) of mass $2m$.



6. A 1 m long tube is open at one end and closed at the other end. Initially it contains a liquid column of 55 cm which traps 25 cm column of air when held horizontal. Now the tube is held vertical with open end at bottom. Now what will be the equilibrium height of liquid column (in cm), after fall of some liquid, if some liquid falls? (Neglect surface tension, assume temperature of the gas remains constant). $P_{\text{atm}} = 10^5 \text{ Pa}$, $\rho_{\text{liquid}} = 10^2 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$.

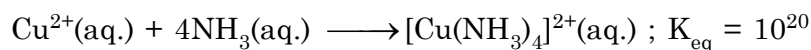
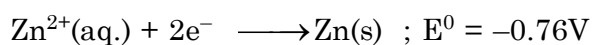
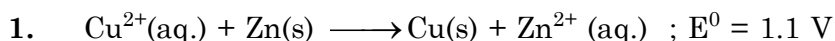


Space for Rough Work

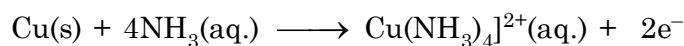
PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

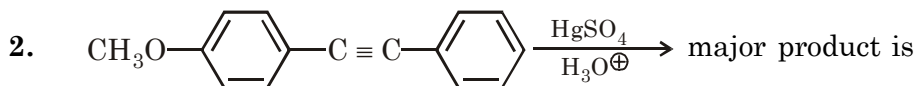
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If $\frac{2.303RT}{F} = 0.06$ then the E^0 of following reaction is -



- (A) 0.34 V (B) 0.76 V (C) 0.26 V (D) 0.14 V



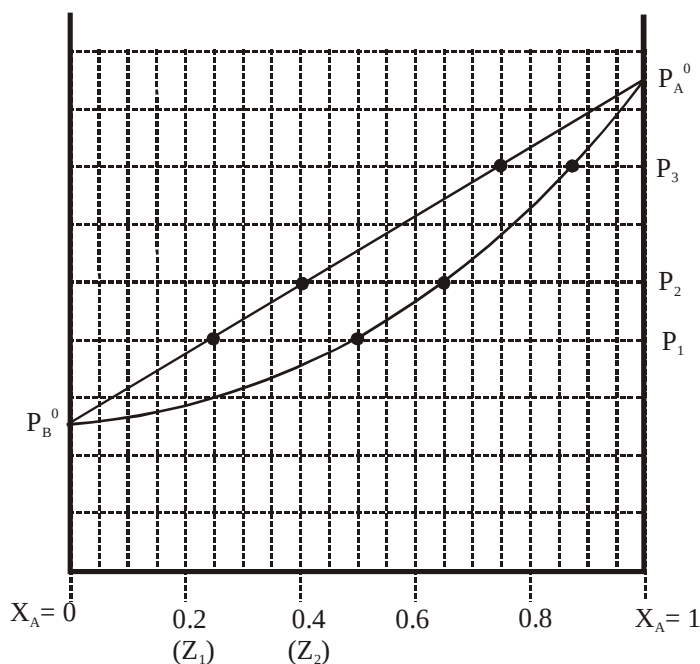
- (A) $\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{CH}_2-\text{C}_6\text{H}_5$ (B) $\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\text{CH}_2-\text{C}(=\text{O})-\text{C}_6\text{H}_5$
 (C) $\text{HO}-\text{C}_6\text{H}_4-\text{CH}_2-\text{C}(=\text{O})-\text{C}_6\text{H}_5$ (D) $\text{HO}-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{CH}_2-\text{C}_6\text{H}_5$

3. Which of the following having highest density among the following?

- (A) Zn (B) Co (C) V (D) Cr

Space for Rough Work

4. With the help of following phase diagram, select the only **INCORRECT** statement -



- (A) Mole fraction of B in vapour at pressure P_1 for solution Z_2 is 0.5
 (B) Mole fraction of A in liquid at pressure P_3 for solution Z_2 is 0.4
 (C) Mole fraction of B in liquid at pressure P_1 for solution Z_2 is 0.75
 (D) Mole fraction of A in liquid at pressure P_2 for solution Z_1 is 0.4

5. $\text{CH}_3 - \text{CH}_2 - \text{C} \equiv \text{CH} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{Cl} \xrightarrow[\text{BaSO}_4]{\text{H}_2, \text{Pd, (excess)}} (\text{X}) \xrightarrow[\text{HOH}]{\text{CH}_3\text{MgBr}} \text{Y} + \text{Z}$
 Products Y and Z are

- (A) Enantiomeric pair of
 (B) Diastereomeric pair of
 (C) Enantiomeric pair of
 (D) Product (Y) and (Z) are not stereoisomers

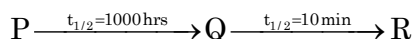
6. What is the overall complex dissociation equilibrium constant for the $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ion given that β_4 for the complex is 2.0×10^{13} .
 (A) 2.0×10^{13} (B) 5×10^{-14} (C) 2×10^{-13} (D) data not sufficient

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

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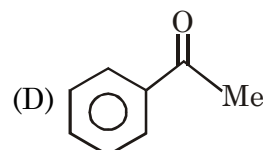
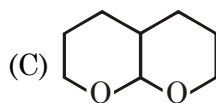
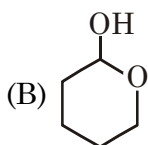
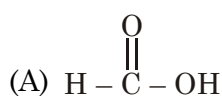
7. Radionuclide 'P' decays into 'Q' and 'Q' decays into 'R' as :



Number of nuclei of Q and R after 1000 hr, if we start with 2 mol P, are.

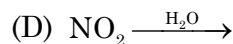
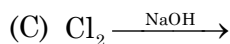
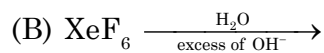
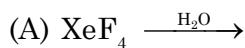
$$[N_A = 6 \times 10^{23}]$$

- (A) $N_Q = 6 \times 10^{23}$ (B) $N_Q = 10^{20}$ (C) $N_R = 6 \times 10^{23}$ (D) $N_R = 10^{23}$
8. Which of the following compounds can show positive test with tollen's reagent?



Space for Rough Work

9. Identify disproportionation reaction(s).



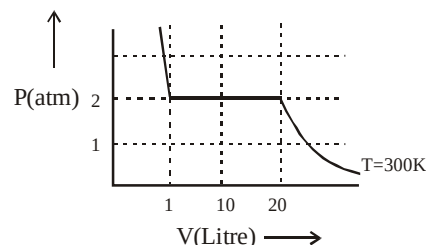
10. A 1kg real gas is liquified at temperature 300K as shown in diagram. Select the correct statement-

(A) 2 atm is vapour pressure of liquid at 300 K

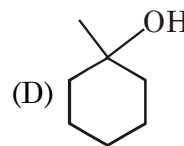
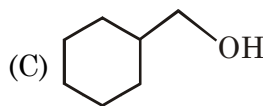
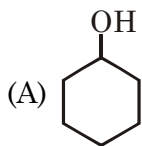
(B) Density of gas when V equals to 10 litre, is 0.1 gm / ml

(C) Density of gas when V equals to 10 litre, is 0.05 gm / ml

(D) Density of liquid when V equals to 10 litre, is 1 gm / ml



11. Which of the following alcohols can change the colour of acidic dicromate solution :



12. Which of the following statement(s) is/are incorrect?

(A) Hydrometallurgy is used to recover Cu from Cu scraps

(B) Hydrometallurgy is used for extraction of Ag from their native ore

(C) Na_2CO_3 is used as leaching agent in Hall's Heroult method

(D) Cupellation is used to remove Pb from impure Ag/Au

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

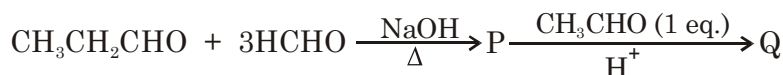
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A perfect diatomic gas at 500K undergoes following steps of process-
 (a) An isochoric change of state from (0.5 atm, 80 litre) to (1atm, 80 litre)
 (b) An isobaric change from (1atm, 80 litre) to (1atm, 20 litre).

If all processes are irreversible then calculate the magnitude of entropy change (in J/K) of surrounding for entire process.

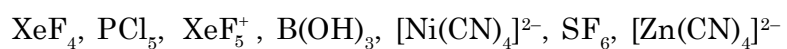
[Use: $\ln 2 = 0.7$, $R = 0.08$ atm-litre/K-mole, 1lit-atm = 100 J.]

2. Formation of (P) is known as Tollen's reaction. Identify total number of stereoisomers for 'Q'

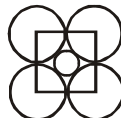


Space for Rough Work

3. Find the number of chemical species having maximum number of atoms may lie in one plane is 5.



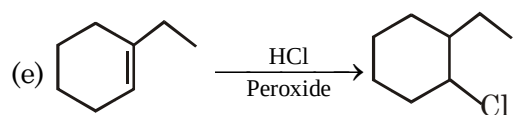
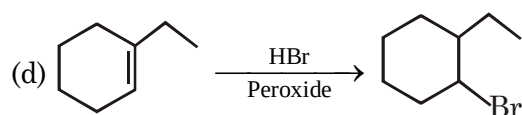
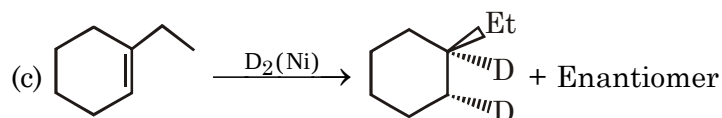
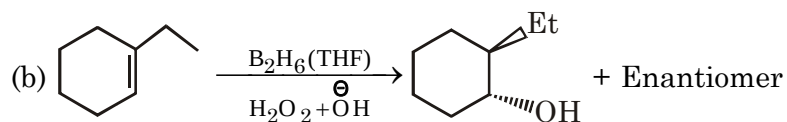
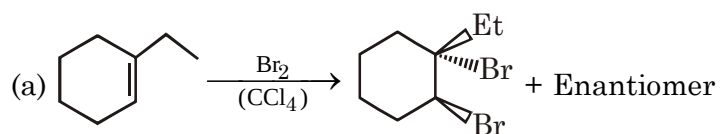
4. Packing fraction of two dimensional unit cell shown in figure is



[Take $\pi = 3.2$, $\sqrt{2} = 1.4$]

Space for Rough Work

5. How many reactions are written with their correct product?



6. Find the number of elements from the given set have greater tendency to convert in M^{+2} to M^{+3} than Cr.

Ti, Fe, V, Mn, Co

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 18)**

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

-
1. If ABCD is regular tetrahedron with length of any edge ' ℓ ', then the minimum distance of any vertex from the opposite face is -
- (A) $\sqrt{\frac{2}{3}}\ell$ (B) $\sqrt{\frac{3}{2}}\ell$ (C) $\frac{2}{3}\ell^2$ (D) $\frac{1}{\sqrt{3}}\ell$
2. If $f(x)$ is continuous and differentiable over $[-2, 5]$ and $-4 \leq f'(x) \leq 3$ for all x in $(-2, 5)$, then the greatest possible value of $f(5) - f(-2)$ is -
- (A) 7 (B) 9 (C) 15 (D) 21
3. Area bounded by $f(x) = \min(\{x - 2\}, \{3 - x\})$, $x = 4$, $x = -4$ and x -axis is equal to (where $\{.\}$ denotes fractional part)
- (A) 1 (B) $3/2$ (C) 2 (D) 4
-

Space for Rough Work

4. Circles $C_1 : x^2 + y^2 = 9$ and $C_2 : x^2 + y^2 - 4x = 0$. Common chord PQ of C_1 & C_2 intersects the direct common tangent at R. If $RP + RQ = \sqrt{\frac{a}{b}}$, where a & b are relatively prime. Then (a + b) equals-
- (A) 65 (B) 63 (C) 71 (D) 79
5. Let $\vec{a} = \hat{i} + 3\hat{j} + 5\hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} + 7\hat{k}$, $\vec{c} = \hat{i} + \hat{j} - \hat{k}$. If the volume of parallelopiped formed by the vectors $2\vec{a} + \vec{b} + \vec{c}$, $\vec{a} - \vec{b} + 2\vec{c}$ and $\vec{a} + 3\vec{b} + \vec{c}$ is v, then
- (A) $v \in [8, 25]$ (B) $v \in (25, 50)$ (C) $v \in [50, 75]$ (D) $v > 75$
6. The equation of the parabola whose focus is the point (0, 0) and the tangent at the vertex is $x - y + 1 = 0$, is
- (A) $x^2 + y^2 - 2xy - 4x + 4y - 4 = 0$ (B) $x^2 + y^2 - 2xy + 4x - 4y - 4 = 0$
- (C) $x^2 + y^2 + 2xy - 4x + 4y - 4 = 0$ (D) $x^2 + y^2 + 2xy - 4x - 4y + 4 = 0$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. If $\phi(x) = 3f\left(\frac{x^2}{3}\right) + f(3-x^2) \quad \forall x \in (-3,4)$ where $f''(x) > 0 \quad \forall x \in (-3,4)$, then $\phi(x)$ is -

(A) increasing in $\left(\frac{3}{2}, 4\right)$

(B) decreasing in $\left(-3, -\frac{3}{2}\right)$

(C) increasing in $\left(-\frac{3}{2}, 0\right)$

(D) decreasing in $\left(0, \frac{3}{2}\right)$

Space for Rough Work

8. Consider two sets $A = \{1, 2, 3\}$ & $B = \{1, 2, 3, 4, 5\}$. A number is selected each from set A & set B. Let p denotes the probability that the selected number from set A is not more than the selected number from set-B & q denotes the probability that the number from set-B is not more than the number from set A, then -
- (A) $p = 2q$ (B) $2p = 3q$ (C) $p + q = \frac{4}{5}$ (D) $p + q = \frac{6}{5}$
9. $P(x)$ is a least degree polynomial such that $(P(x) - 1)$ is divisible by $(x - 1)^2$ & $(P(x) - 3)$ is divisible by $(x + 1)^2$, then
- (A) graph of $y = P(x)$ is symmetric about origin
 (B) $y = P(x)$ has two points of extrema
 (C) $\int_{-4}^{\lambda} P(x)dx = 0$ for exactly one value of λ .
 (D) $\int_{-4}^{\lambda} P(x)dx = 0$ for exactly two values of λ .
10. If $z_1, z_2, z_3 \in \mathbb{C}$ and satisfying $|2z_1 - 1| = 4$, $|2z_2 - 5| = 1$ and $|z_3 - 3| = 1$ respectively, then
- (A) least value of $|z_1 + z_2 + z_3|$ is $\frac{3}{2}$ (B) least value of $|z_1 + z_2 + z_3|$ is $\frac{5}{2}$
 (C) greatest value of $|z_1 + z_2 + z_3|$ is $\frac{19}{2}$ (D) greatest value of $|z_1 + z_2 + z_3|$ is $\frac{11}{2}$

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11. Equation of curve passing through (3, 4) and satisfying the differential equation,

$$y \left(\frac{dy}{dx} \right)^2 + (x - y) \frac{dy}{dx} - x = 0 \text{ can be}$$

- (A) $x^2 + y^2 = 25$ (B) $x^2 - y^2 + 7 = 0$ (C) $xy = 12$ (D) $x - y + 1 = 0$

12. Let $f(x) = \begin{cases} e^{\tan^{-1}(x-1)} + (x-1)^{2017} \sin^2 \frac{1}{(x-1)} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$ and $\ell = \lim_{n \rightarrow \infty} \left(n \sum_{k=1}^{10} f\left(1 + \frac{k}{n}\right) - 10n \right)$

- (A) $f(x)$ is continuous but non-derivable at $x = 1$
 (B) $f(x)$ is differentiable at $x = 1$
 (C) $55f'(1) - \ell = 0$
 (D) ℓ is divisible by 11.

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
●	●	●	●
①	①	①	①
②	②	②	②
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let $\vec{x}, \vec{y}$ are non zero and non collinear vectors such that $(a\alpha^2 + b\alpha + c)\vec{x} + (a\beta^2 + b\beta + c)\vec{y} + (a\gamma^2 + b\gamma + c)(\vec{x} \times \vec{y}) = 0$, then $(a + b + c)$ is (where $a, b, c \in \mathbb{R}$)
- Let $P(x)$ be a polynomial with degree 4 such that $\lim_{x \rightarrow 0} \left(\frac{P(x)}{x} - 12 \right) = 0$. The graph of $P(x)$ passes through $A(1, 25)$ & $B(-1, -7)$ & has extremum at point B. If the equation $|P(x)| = k$ has exactly 3 roots, then k is

Space for Rough Work

3. Consider $I_1 = \int_0^{\pi/2} (\cos x)^{\sqrt{17}+3} dx$ and $I_2 = \int_0^{\pi/2} (\cos x)^{\sqrt{17}+5} dx$, then $\frac{I_1}{I_2}$ is equal to (Use $\sqrt{17} = 4.12$).
4. Slope of normal to the ellipse at a point P is $\frac{3}{4}$ and eccentricity of ellipse is $\frac{1}{3}$. If this normal makes acute angle ' β ' with its focal chord through P, then $\sin\beta$ is
5. Let A is a four order diagonal matrix whose entries are complex numbers such that $A^4 = I_4$. If trace of A is zero, then number of such matrices is
(Here I_4 is four order unit matrix)
6. Consider a sequence of distinct terms $\{a_n\}$ with $a_1 = 2$ & $a_n = \frac{a_{n-1}^2}{a_{n-2}}$ for $\forall n \geq 3$. If $a_5 \leq 162$ and a_2, a_5 are positive integers than sum of possible value(s) of a_5 is

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QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator